Dr Andi Lowe

Data Scientist · PhD in Particle Physics · Ex-CERN · Breakthrough Prize Laureate

Budapest, Hungary • dr.andi.lowe@gmail.com • +36 20 295 6795 • linkedin.com/in/drandilowe

Professional Profile

- Machine Learning-focused Data Scientist with 8+ years of industry experience working with messy, real-world data, delivering analytical and ML-driven solutions in regulated and technically complex environments.
- Background in Python and R development, performance optimisation, and algorithmic problem-solving, with experience spanning high-performance computing, statistical modelling, and applied machine learning. Strong focus on data quality assessment, feature engineering, and rigorous model validation.
- PhD-qualified with extensive experience in low-latency, high-throughput inference pipelines and managing massive real-time data streams (*i.e.*, "Big Fast Data").
- Proven consultant and stakeholder manager, adept at translating complex analytical results into clear, actionable insights for senior business stakeholders across diverse domains.
- Former CERN researcher and member of the team that discovered the Higgs boson.
- British citizen with unrestricted right to work in Ireland; immediately available to relocate.

Work History

Data Scientist (Client-Facing Consulting Role) — *Budapest, Hungary*
EPAM Systems Inc. — *June 2017 – present*

Led end-to-end data science and applied machine learning engagements for global clients, working independently across the full analytics lifecycle, from problem definition and modelling through validation, documentation, and delivery of decision-support solutions in business and operational settings.

- Architected, validated, and documented predictive models on large, noisy time-series datasets using AWS SageMaker, Python, and SQL to identify failure risk and operational anomalies in GPS fleet-tracking data. Reduced false fault detection by **~85%**. Delivered actionable insights to operations teams; work resulted in a patent filing *(Verizon Connect)*.
- Built a proof of concept for remote sensing using millimetre-wave radar technology, performing signal analysis and image processing with OpenCV *(Texas Instruments)*.
- Designed and implemented machine learning models in Python (scikit-learn) to detect clinically relevant behavioural patterns from wearable sensor data, supporting treatment monitoring by healthcare professionals. Achieved **87%** average gesture-recognition accuracy (**79–100%** per gesture), validating feasibility and enabling progression to MVP. Provided technical leadership by directing data collection strategy and advising on hardware selection to align analytical and practical constraints *(Child Mind Institute)*.
- Performed end-to-end analysis of neuromarketing data to identify key drivers and inform campaign strategy. Presented insights directly to senior business stakeholders *(Merck Sharp & Dohme)*.
- Led the design and implementation of privacy-preserving data pipelines, in Python and R with Docker, ensuring GDPR compliance and protecting a critical **~$250 million** annual revenue stream. Supported data governance in a highly regulated environment. Research and consultancy to help define corporate strategy with regards to reducing the client's digital carbon footprint *(London Stock Exchange Group)*.
- Built extract, transform, and load (ETL) processes and data pipelines using KNIME to prepare data and support financial reporting databases, reducing processing time from over 24 hours to under 2 hours (**12× faster**) *(Bayer AG)*.

Research Fellow — *Budapest, Hungary*
Wigner Research Centre for Physics, Hungarian Academy of Sciences — *Sept 2013 – May 2017*

Conducted data analysis, statistical modelling, and machine learning in R and C++ for the ALICE experiment at CERN, which recreates conditions believed to have existed shortly after the Big Bang.

- Devised novel ML algorithms to identify particles based on their decay properties. First use of ML in this experiment.

Postdoctoral Fellow and Deputy Team Leader — *Fresno, CA, USA*
California State University, Fresno — *Apr 2010 – Oct 2012*

Performed feature engineering in C++ for data-mining analyses using Monte Carlo-generated synthetic data to improve search sensitivity for new particles and reduce the time and cost of experimental data collection.

- Contributed to National Science Foundation (NSF) grant proposals totalling over **$1.6 million** in research funding.
- Stationed at CERN, Geneva, Switzerland.

Postdoctoral Fellow *Bloomington, IN, USA*
Indiana University *Feb 2008 – Aug 2009*
Developed and productionised high-performance C++ and Python algorithms for real-time data processing at input rates of **~100 GB/s**, operating in distributed computing environments to process tens of petabytes of data for the ATLAS experiment.

- Stationed at CERN, Geneva, Switzerland.

Education

Royal Holloway, University of London *Egham, UK*
Doctorate (PhD) in Particle Physics
Designed and optimised high-performance C++ software for real-time, multi-stage data reduction pipelines, managing raw ingestion rates of **60 TB/s**.

- Engineered and deployed software for a cascade classifier system that achieved a **200,000× data reduction** (300 MB/s output), enabling the sustainable permanent storage of petabyte-scale datasets.
- Performed deep-level CPU and time profiling, memory optimisation, and algorithmic refactoring of core software and devised improvements that made it **8 times faster**, thus meeting a critical requirement of the system.
- Analysed petabyte-scale datasets using distributed computing systems.
- Engineered, tested, and deployed software used in the discovery of the Higgs boson.
- Stationed at CERN, Geneva, Switzerland for 1.5 years.

Royal Holloway, University of London *Egham, UK*
Master's degree (MSc) in Particle Physics
Investigated the search potential for the Higgs boson via the H to bb (bottom quark-antiquark) decay channel using Monte Carlo-generated synthetic data with the ATLAS detector at CERN.

- Conducted the first data-mining analysis of this type entirely in C++, setting a benchmark for subsequent research.
- Achieved a signal significance of 4.87 sigma (previously 3.66 sigma; 5 sigma is the threshold for discovery).

Royal Holloway, University of London *Egham, UK*
Bachelor's degree (BSc) in Physics

Leadership and Consulting Skills

Communication: Skilled at collaborating with clients as a consultant and effectively communicating with business stakeholders and technical teams to convey complex data-driven insights and technical specifications. Proficient in translating client requirements into detailed project documentation. Invited speaker at numerous international conferences. Experienced in visual storytelling and data visualisation best practices. Adept at simplifying complex technical concepts for diverse audiences. Extensive experience working with remote, distributed teams.

Problem solving: Proven ability to lead independent research, analyse complex problems, and develop innovative, effective solutions.

Project management: Experienced in managing multiple projects under strict deadlines, both onsite and remotely. Familiar with Agile and Waterfall software development methodologies.

Professional Certificates

- **AWS Certified AI Practitioner**, Amazon Web Services (AWS), issued Nov. 2024.
- **Salesforce Certified AI Associate**, Salesforce, issued Jan. 2025.

Computing Skills

Machine learning frameworks/tools: Python-based ML tooling (Scikit-Learn, XGBoost, LightGBM, Keras, DBSCAN, HDBSCAN, SHAP values), AWS Sagemaker, KNIME, H2O, ROOT (CERN's C++ data analysis framework).

Programming languages: Python (NumPy, Pandas, Polars, Numba, etc), SQL, R, C++, Bash, Octave.

Data visualisation: Matplotlib, Seaborn, Plotnine, ggplot2, Plotly, TIBCO Spotfire, Tableau, Flexdashboard.

Notebooks/Documentation: Jupyter Notebooks, Google Colab, R Markdown, Confluence, LaTeX, Doxygen.

Software development: Docker, Git/GitHub/GitLab, Jira, RStudio, UML, Valgrind, Visual Studio Code.

Operating systems: Unix, Linux, Microsoft Windows.

Other: object-oriented analysis and design, CPU and time profiling, code optimisation, memory debugging.